Prompt Probes Read Solve Rate, Not Robustness: Negative Results from the AIMO Interpretability Challenge

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`https://github.com/byeolki/aimo-interp`

Abstract

We predict, from one forward pass over the prompt, whether a small reasoning model solves an olympiad problem robustly (Small Models track). A linear probe looked strong on the official training rows (balanced accuracy up to 0.78), but those rows never give different labels to different models on the same problem, so it can succeed by reading problem difficulty. We therefore labeled 150 new problems for three of the four models ourselves. A model's hidden state predicts its own solve rate (r = 0.69 to 0.75), better than the agreement of four sampled answers (r = 0.53), but not its robustness: with model identity removed, self-probes are at chance (0.42 to 0.49), and their apparent gains come from model-level priors that the official evaluation neutralizes. A cheap test-time counterfactual check reaches 0.64 to 0.66 on our labels but 0.38 to 0.50 on the organizers'. We document two pitfalls that silently corrupt self-generated labels and release all code and data.

1 Setting

For a model m and problem q, the task is to predict whether m's accuracy on q survives perturbations that keep the deductive chain [1]. The Small Models track covers Qwen3.5-4B, Skywork-OR1-Math-7B, Olmo-3-7B-Think and DeepSeek-R1-0528-Qwen3-8B, evaluated offline in 3600 s on one RTX PRO 6000. Public supervision is 34 labeled rows over 20 problems. Validation and test sets are balanced per problem and per model; always-true scores exactly 0.50 on the 14-case validation set.

2 A shared prompt probe reads difficulty

We fit L2 logistic probes on every layer's last-token and mean-pooled prompt states with problem-grouped 5-fold CV (3 seeds). A *shared* probe uses one encoder for all targets plus a target one-hot; a *self* probe uses the target's own states (Figure 1).

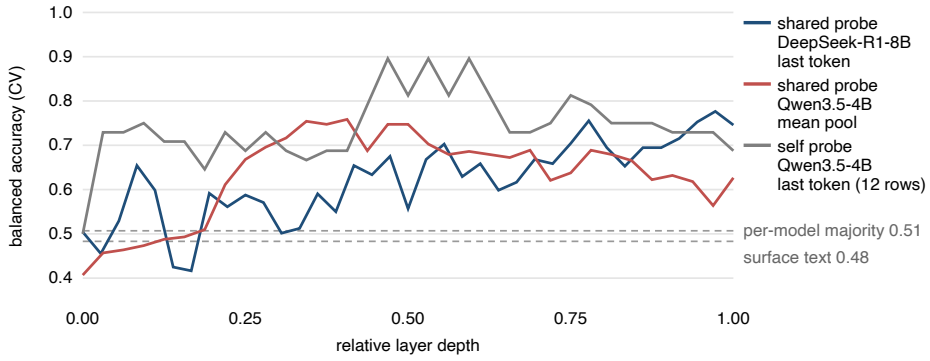


Figure 1. Balanced accuracy on the 34 official rows by relative depth. Surface text features and the per-model majority stay at chance.

The rising DeepSeek curve (0.72 over the top quarter of layers) is not evidence for the target: of the 9 problems labeled for more than one model, none has differing labels, and the one-hot enters only as a bias, so the probe cannot separate models on one problem. Submitted, it scored 0.643 (9/14).

3 Self-generated labels

We labeled 150 integer-answer problems absent from the official data (AIME 2022 to 2026, AMC 12 2022 to 2023, HMMT, BRUMO, SMT, CMIMC) with the organizers' sampling settings (T = 1.0, top-k 40, top-p 0.95): 8 samples on the original and, if solved at least once, 6 on each of three rule-based variants (rename, typos, distractor). Robust if the largest family drop is at most 0.10, spurious if at least 0.25; never-solved problems count as robust. Olmo was dropped (5 concurrent sequences on a 32 GB GPU). Two pitfalls nearly corrupted the labels. **Truncation:** at a 12k-token cap 90% of Qwen3.5-4B samples ended without an answer (mean accuracy 0.10); capping reasoning at 8k and forcing an answer (`</think> ... \boxed{`) gave 0.49. **fp8 KV cache:** in vLLM 0.31 Skywork looped on single tokens (accuracy 0.12 on finished answers); bf16 gave 0.98. The result is 393 labels; 44 problems (117 rows) have labels that differ across models.

4 Self-probes: solve rate yes, robustness no

Method	mixed problems (117)	official rows (25)
per-model majority	0.54	0.68
shared probe (Section 2)	0.58	0.68
self probe, direct	0.61	0.57
self probe, solve rate then robustness	0.61 to 0.63	0.45 to 0.51
self probe, blend (submitted)	0.63	0.64
sampled base accuracy (reference)	0.74	n/a

Table 1. Out-of-fold accuracy (problem-grouped, 3 seeds). One standard error on 117 rows is about 4.5 points.

A ridge probe on a model's own last-token state predicts its 8-sample solve rate at r = 0.69 to 0.75, whereas the majority agreement of four sampled answers correlates with four other samples' solve rate at only r = 0.53. Robustness is another matter. Drawing one robust and one spurious model per mixed problem (Figure 2), the model prior alone gives 0.68 and ranking by raw probe score gives no more (0.63 to 0.69); with scores standardized per model, or thresholded at each model's median, the probe falls to 0.42 to 0.49. Its edge over the shared probe is a model-identity prior, which the official sets remove; on Codabench it also scored 0.643.

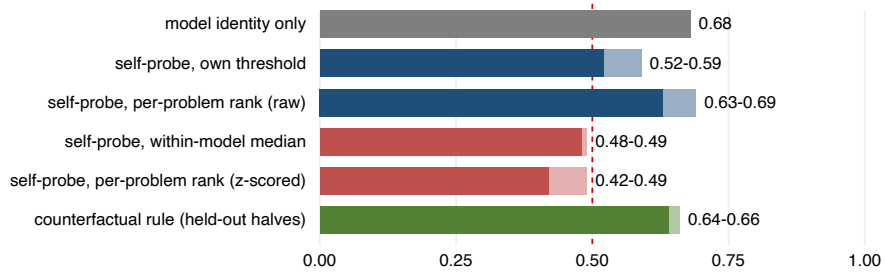


Figure 2. Accuracy on per-problem balanced draws of our labels (chance dashed; ranges over seeds). Red rules remove model identity.

5 A test-time counterfactual check does not transfer

We then measured behaviour at test time: the target answers the original twice and each variant once under a time-managed budget (8k down to 1k tokens), and a problem is robust if no variant loses more than 0.1 of the model's own majority answer.

Evaluation	counterfactual	reference
our labels, balanced (held-out halves)	0.64 to 0.66	self probe 0.42 to 0.49
34 official rows (RTX 5090, 3072 s)	0.38 (bal. 0.48)	always-true 0.68
Codabench validation (14)	0.50	self probe 0.643

Table 2. The same rule against our labels and the organizers' labels.

It predicts only 9 of 34 official rows robust (70% are). Five samples let one unstable answer flip the call, while the official labels use more samples, budgets up to 100k tokens and LLM-written perturbations; agreement with our labels came from sharing our choices.

6 Discussion and limitations

Prompt representations encode whether a model can solve a problem, not whether its solution survives perturbation. Validating probes only on rows without per-problem label variation, or without a model-identity baseline, hides this, and self-generated labels inherit their generator's budget, perturbations and sample counts. Limitations: 150 problems, three models, an 8k budget, rule-based perturbations, the never-solved-is-robust convention, and a 14-case leaderboard. The final entry (`submissions/self_probe`) runs the 34 official rows offline in 38 s; labels, generations and all scripts are in the repository (about USD 10 of rented GPU time in total).

References

[1] AIMO Interpretability Challenge. Competition proposal, NeurIPS 2026 Competition Track. arXiv:2607.13899, 2026. https://aimo-interp.github.io/